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Unit 1

Basic Concepts of Chemistry and Chemical Calculations

3PJYME

We think there is colour, we think there is sweet, we think there is bitter, but in reality there are atoms and a void.

– Democritus



Natural Fibre (Cotton)

Synthetic Fibre (Nylon)





Pesticide (Malathion)

Pharmaceutical (Tablets)





Refinery (Oil)

Learning Objectives

After studying this unit, students will be able to

- explain the importance of chemistry in different spheres of life.
- classify different substances into elements, compounds and mixtures.
- define atomic mass and molecular mass.
- define the amount of substance using SI unit 'mole'.
- describe Avogadro number.
- explain the relationship among mass, moles and number of atoms (or) molecules and perform calculations relating to the conversions.
- define equivalent mass and calculate equivalent mass of acid, base and oxidising/reducing agents.
- deduce empirical and molecular formula of a compound from experimental data.
- solve numerical problems based on stoichiometric calculations.
- identify the limiting reagent and calculate the amount of reactants and products in a reaction.
- define the terms oxidation, reduction, oxidant and reductant.
- predict the oxidation states of elements in various compounds.
- explain the process involved in a redox reaction and describe the electron transfer process.
- classify redox reactions into different types.
- formulate a balanced redox reaction from two half-reactions.